

Ex. No. 8	PERFORMING DATA TRANSFORMATION
Date: 09-03-2026	

Aim:

To perform a Data Transformation operation on Vehicle Maintenance Management System.

Vehicle Maintenance collection:

```
db.maintenance.insertMany([
  { vehicleName: "Honda City", serviceType: "Oil Change", serviceCount: 2, serviceCost: 3000 },
  { vehicleName: "Hyundai i20", serviceType: "Brake Service", serviceCount: 5, serviceCost: 5000 },
  { vehicleName: "Toyota Innova", serviceType: "Engine Check", serviceCount: 3, serviceCost: 7000 },
  { vehicleName: "Kia Seltos", serviceType: "Battery Service", serviceCount: 4, serviceCost: 4000 },
  { vehicleName: "Mahindra XUV700", serviceType: "Full Service", serviceCount: 1, serviceCost: 9000 },
  { vehicleName: "Maruti Swift", serviceType: "Tyre Replacement", serviceCount: 6, serviceCost: 2000 },
  { vehicleName: "Hyundai Creta", serviceType: "General Service", serviceCount: 7, serviceCost: 3500 },
  { vehicleName: "Tata Nexon", serviceType: "AC Service", serviceCount: 8, serviceCost: 4500 },
  { vehicleName: "BMW X1", serviceType: "Premium Service", serviceCount: 2, serviceCost: 15000 },
  { vehicleName: "Audi A4", serviceType: "Engine Service", serviceCount: 1, serviceCost: 18000 }
])
```

Bucket (group vehicles based on service cost):

```
db.maintenance.aggregate([
  {
    $bucket: { groupBy: "$serviceCost",
    boundaries: [0, 3000, 6000, 12000, 20000],
    default: "Other", output: { count: { $sum:
    1 }, vehicles: { $push: "$vehicleName" }
    }
  }
])
```

Output:

```
{ _id: 0,  
  count: 1,  
  vehicles: [  
    'Maruti Swift'  
  ]  
}  
  
{ _id:  
  3000,  
  count: 5,  
  vehicles: [  
    'Honda City',  
    'Hyundai i20',  
    'Kia Seltos',  
    'Hyundai Creta',  
    'Tata Nexon'  
  ]  
}  
  
{ _id:  
  6000,  
  count: 2,  
  vehicles: [  
    'Toyota Innova',  
    'Mahindra XUV700'  
  ]  
}  
  
{  
  _id: 12000,  
  count: 2, vehicles:  
  [  
    'BMW X1',  
    'Audi A4'
```

```
]
```

```
}
```

Facet (multiple aggregations in one):

```
db.maintenance.aggregate([
```

```
{
```

```
$facet: {
```

```
"ByServiceType": [
```

```
{
```

```
$group: { _id:
```

```
"$serviceType",
```

```
totalVehicles: { $sum: 1 } 
```

```
} 
```

```
} 
```

```
],
```

```
"HighServiceCount": [
```

```
{
```

```
$match: { serviceCount:
```

```
{ $gte: 5 } 
```

```
} 
```

```
},
```

```
{
```

```
$project: { vehicleName:
```

```
1, serviceCount: 1,
```

```
_id: 0
```

```
} 
```

```
} 
```

```
]
```

```
} 
```

```
} 
```

```
])
```

Output:

```

{
  ByServiceType: [
    { _id: 'Oil
      Change',
      totalVehicles: 1
    },
    { _id: 'Brake
      Service',
      totalVehicles: 1
    },
    { _id: 'Engine
      Check',
      totalVehicles: 1
    },
    { _id: 'Battery
      Service',
      totalVehicles: 1 }
  ],
  HighServiceCount: [ {
    vehicleName: 'Hyundai i20',
    serviceCount: 5
  }, { vehicleName: 'Maruti
    Swift', serviceCount: 6
  }, { vehicleName: 'Hyundai
    Creta', serviceCount: 7
  }, { vehicleName: 'Tata
    Nexon', serviceCount: 8
  }
  ]
}

```

Sort by service cost (high to low):

```
db.maintenance.find().sort({ serviceCost: -1 })
```

Output:

```
{ vehicleName: 'Audi A4',  
  serviceType: 'Engine Service',  
  serviceCount: 1, serviceCost:  
    18000  
}  
  
{  
  vehicleName: 'BMW X1',  
  serviceType: 'Premium Service',  
  serviceCount: 2, serviceCost:  
    15000  
}  
  
{ vehicleName: 'Mahindra  
XUV700', serviceType: 'Full  
Service', serviceCount: 1,  
  serviceCost: 9000  
}
```

```
{ vehicleName: 'Toyota  
Innova', serviceType: 'Engine  
Check', serviceCount: 3,  
  serviceCost: 7000
```

```
} { vehicleName: 'Hyundai  
i20', serviceType: 'Brake  
Service', serviceCount: 5,  
  serviceCost: 5000
```

```
} Skip first 2 documents:
```

```
db.maintenance.find().skip(2)
```

Output:

```
{ vehicleName: 'Toyota  
Innova', serviceType: 'Engine
```

```
Check', serviceCount: 3,
serviceCost: 7000
} { vehicleName: 'Kia Seltos',
serviceType: 'Battery Service',
serviceCount: 4, serviceCost:
4000
}
{ vehicleName: 'Mahindra
XUV700', serviceType: 'Full
Service', serviceCount: 1,
serviceCost: 9000
} { vehicleName: 'Maruti Swift',
serviceType: 'Tyre Replacement',
serviceCount: 6, serviceCost:
2000
}
```

Limit to 5 documents: db.maintenance.find().limit(5)

Output:

```
{ vehicleName: 'Honda
City', serviceType: 'Oil
Change', serviceCount: 2,
serviceCost: 3000
} { vehicleName: 'Hyundai
i20', serviceType: 'Brake
Service', serviceCount: 5,
serviceCost: 5000
} { vehicleName: 'Toyota
Innova', serviceType: 'Engine
Check', serviceCount: 3,
serviceCost: 7000
} { vehicleName: 'Kia Seltos',
serviceType: 'Battery Service',
```

```
serviceCount: 4, serviceCost:
```

```
4000
```

```
}
```

```
{ vehicleName: 'Mahindra
```

```
XUV700', serviceType: 'Full
```

```
Service', serviceCount: 1,
```

```
serviceCost: 9000
```

```
}
```

Result:

Thus, the above queries to perform Data Transformation on the 'Vehicle Maintenance' collection in MongoDB have been executed successfully.

